

Combined Oxidative Phosphorylation Deficiency 24 (COXPD24 / NARS2-Related Disease): A Comprehensive Disease Characterization

Disease: Combined Oxidative Phosphorylation Defect Type 24 **MONDO:** MONDO:0014547 · **OMIM:** #616239 · **Gene:** *NARS2* (OMIM 612803; HGNC:24406) · *Category:** Mendelian, autosomal recessive

Summary

Combined Oxidative Phosphorylation Deficiency 24 (COXPD24) is an ultra-rare, autosomal-recessive, multisystem mitochondrial disorder caused by biallelic pathogenic variants in **NARS2**, the nuclear gene encoding the **mitochondrial asparaginyl-tRNA synthetase** (mt-AsnRS). NARS2 charges mitochondrial tRNA-Asn with asparagine, an obligatory step in the intramitochondrial translation of the 13 OXPHOS subunits encoded by mtDNA. When NARS2 function is lost, charged mt-tRNA-Asn falls, mitochondrial protein synthesis fails, and the cell manifests a **combined oxidative phosphorylation deficiency—predominantly of complex I and complex IV**—which is the defining biochemical hallmark of the disease ([PMID: 25385316](#); [PMID: 25807530](#)).

Clinically, COXPD24/NARS2 disease is strikingly **phenotypically variable and genotype-graded**, spanning a continuous spectrum from isolated nonsyndromic sensorineural hearing loss (DFNB94) at the mild end to severe, early-onset Leigh and Alpers-type encephalopathy with refractory epilepsy, developmental delay/intellectual disability, myopathy, spastic paraplegia, and ataxia at the severe end. **Sensorineural hearing loss is a cardinal feature**, and **neonatal/infantile insulin-dependent diabetes** has recently been recognized as an important part of the phenotype, reflecting a critical role for NARS2 in pancreatic β -cells

([PMID: 33596490](#); [PMID: 40887432](#)). Onset is typically neonatal to early childhood; severe encephalopathic forms carry high early mortality, whereas isolated deafness is compatible with long survival.

Diagnosis rests on **molecular genetics (whole-exome/whole-genome sequencing)**, supported by elevated blood/CSF lactate, characteristic MRI (symmetric Leigh-type basal ganglia/thalamus/brainstem lesions, or progressive cerebral atrophy and delayed myelination), and muscle-biopsy demonstration of combined complex I + IV deficiency. **No disease-modifying therapy exists**; management is supportive (antiepileptic drugs, hearing rehabilitation, developmental therapies, nutritional support, insulin for the diabetic subgroup), and prevention is limited to genetic counseling with carrier, prenatal, and preimplantation genetic testing. Substrate (L-asparagine) supplementation is a rational but unproven experimental strategy, and class-level trials of amino-acid supplementation in related mitochondrial aminoacyl-tRNA synthetase disorders have not met efficacy endpoints.

Key Findings

Finding 1 — COXPD24 is caused by biallelic (autosomal-recessive) *NARS2* variants

COXPD24 (OMIM #616239) maps to *NARS2* (OMIM 612803; HGNC:24406) on chromosome 11q14.1, which encodes the mitochondrial asparaginyl-tRNA synthetase. The gene-disease relationship is supported by multiple independent consanguineous and compound-heterozygous families. Vanlander et al. (2015) reported two siblings with a homozygous c.822G>C splice-site variant producing exon-7 skipping and a combined complex I + IV deficiency in skeletal muscle: "The reported variant in *NARS2* results in a combined OXPHOS complex deficiency involving complex I and IV, making *NARS2* a new member of disease-associated aaRS2" ([PMID: 25385316](#)). In the same year, Simon et al. established the broader allelic spectrum—homozygous p.Val213Phe causing nonsyndromic deafness (DFNB94), and compound heterozygous p.Tyr323*/p.Asn381Ser causing Leigh syndrome: "Here we demonstrate association of variants in the mitochondrial asparaginyl-tRNA synthetase *NARS2* with human hearing loss and Leigh syndrome"* ([PMID: 25807530](#)). Biallelic inheritance is confirmed across subsequent case series.

Finding 2 — Mechanism: reduced charged mt-tRNA-Asn impairs mitochondrial translation, and the defect is complementable

The molecular lesion is a reduction in **aminoacylated (charged) mitochondrial tRNA-Asn**. In EBV-transformed lymphoblasts from affected siblings, *"a specific decrease in the amount of charged mt-tRNA(Asn) was demonstrated as compared with controls"* (PMID: 25385316). Downstream, patient Leigh-syndrome fibroblasts show decreased steady-state mt-tRNA-Asn, reduced oxygen consumption rate (OCR), and reduced electron transport chain (ETC) activity. Crucially, these deficits are rescued by wild-type NARS2 but not by the mutant protein: *"we show that a decrease in oxygen consumption rates (OCR) and electron transport chain (ETC) activity can be rescued by overexpression of wild type NARS2"* (PMID: 25807530). Structural mechanisms differ by variant: p.Asn381Ser disrupts NARS2 homodimerization and lowers protein levels, while the truncating p.Tyr323* yields undetectable protein. BN-PAGE in-gel activity confirms decreased complex I and IV activity plus complex V subcomplexes. This complementation experiment provides direct causal proof that the respiratory-chain defect is a consequence of NARS2 loss of function.

Finding 3 — A phenotypically variable multisystem disorder; hearing loss is cardinal, neonatal diabetes newly recognized

Sofou et al. (2021) systematically reviewed ~19 patients and concluded that *"sensorineural hearing impairment is a cardinal feature of early-onset NARS2 associated disease, either isolated or in combination with central nervous system disease"* (PMID: 33596490). Reported phenotypes span nonsyndromic sensorineural hearing loss (DFNB94), Leigh syndrome, Alpers syndrome, infantile-onset refractory epilepsy/epileptic encephalopathy, epilepsia partialis continua, developmental delay/intellectual disability, myopathy, spastic paraplegia, and ataxia. In 2025, Donis et al. expanded the phenotype: among 397 early-onset diabetes patients, 8 carried homozygous NARS2 missense variants [p.(Phe216Leu) ×4, p.(Thr180Asn) ×3, p.(Val440Leu) ×1]; all had insulin-dependent (neonatal) diabetes diagnosed before 6 months (median 4 weeks), 7/8 had low birthweight, 7/8 had epilepsy, and 6/8 had developmental delay. The authors concluded *"NDM is an important feature of COXPD-24 and highlights a critical role for NARS2 in the insulin-secreting pancreatic β -cell"* (PMID: 40887432).

Finding 4 — Genetics: LoF-tolerant nuclear gene; ultra-rare AR disease with genotype-phenotype correlation

gnomAD constraint metrics for NARS2 (ENSG00000137513) show $pLI \approx 0$ and **observed/expected LoF (oe_lof) = 0.95** (90% CI 0.77–1.18), indicating heterozygous loss-of-function is tolerated—exactly as expected for a recessive disease gene where carriers are unaffected. Disease alleles are overwhelmingly biallelic missense or splice variants, frequently homozygous in consanguineous families: p.Pro214Leu (recurrent Scandinavian; PMID: 33596490), p.Ile182Lys (Iranian; PMID: 34374940), p.His167Arg (Turkish; PMID: 36661119), c.822G>C splice (PMID: 25385316), and p.Val213Phe (DFNB94; PMID: 25807530). Severity cosegregates with the biochemical/structural impact of the variant: *"The severity of the genetic lesions and their effects on NARS2 protein structure cosegregate with the phenotype"* (PMID: 25807530).

Finding 5 — Treatment is supportive; substrate supplementation is experimental and unproven

No disease-modifying therapy or approved drug exists. Management is symptomatic: antiepileptic drugs for seizures, hearing aids/cochlear implants, developmental/physical/occupational therapy, nutritional support, and insulin for the neonatal-diabetes subgroup. Class-level evidence from related mitochondrial aminoacyl-tRNA synthetase (mt-ARS) disorders is discouraging: a 2-year open-label pilot of substrate amino-acid supplementation in AARS2/DARS2 leukoencephalopathy was *"safe and well tolerated by all patients, but efficacy endpoints were not met as no significant improvements were observed in global, cognitive, or motor scores"* (PMID: 41075682). A broad aaRS review notes *"amino acid supplementation and dietary interventions have shown effect in select cases, while gene therapy is being explored for dominant ARS-related neuropathies"* (PMID: 42028791). By analogy the rational (unproven) substrate for NARS2 is **L-asparagine (CHEBI:22653)**. Because COXPD24 is a mitochondrial disorder, **valproic acid should be avoided or used cautiously** (PMID: 36661119).

Finding 6 — Anatomy & subcellular localization

UniProt Q96I59 describes NARS2 as a 477-amino-acid **class II aminoacyl-tRNA synthetase localized to the mitochondrial matrix**, with an N-terminal mitochondrial transit peptide (residues 1–14). It catalyzes the ATP-dependent, two-step attachment of asparagine to mt-tRNA-Asn ($\text{Asn} + \text{ATP} \rightarrow \text{Asn-AMP} \rightarrow \text{charged tRNA}$). Disease targets multiple tissues: **CNS** (basal ganglia, thalamus, brainstem in Leigh syndrome; cortex/global atrophy and delayed myelination in Alpers-like disease), **inner ear/cochlea** (sensorineural hearing loss), **skeletal**

muscle (myopathy with combined complex I/IV deficiency), and **endocrine pancreas/ β -cells** (neonatal diabetes). CNS and cochlear involvement is typically bilateral/symmetric: *"Leigh syndrome, which is a neurodegenerative disease characterized by symmetric, bilateral lesions in the basal ganglia, thalamus, and brain stem"* (PMID: 25807530).

Finding 7 — Diagnosis

Definitive diagnosis is genetic. Supporting evidence includes: (1) **biochemistry**—elevated serum/CSF lactate and myocardial/muscle enzymes; (2) **muscle biopsy**—combined complex I + IV deficiency by spectrophotometry and BN-PAGE, plus complex V subcomplexes (*"a combined complex I and IV deficiency in skeletal muscle"*, PMID: 25385316); (3) **neuroimaging**—Leigh-type symmetric lesions or Alpers-like cerebral atrophy with delayed myelination, and epileptiform EEG/epilepsia partialis continua; (4) **molecular genetics**—trio-based WES or WGS identifying biallelic NARS2 variants, with rapid WES shortening the diagnostic odyssey (*"Rapid whole exome sequencing was performed revealing novel biallelic variants in NARS2"*, PMID: 30327238); and (5) **functional confirmation**—decreased charged mt-tRNA-Asn, rescue assays, and LC-MS/MS aminoacylation assays to classify variants of uncertain significance.

Finding 8 — Temporal course & prognosis

Onset is usually neonatal to early childhood (e.g., 9-month-old with status epilepticus, PMID: 41426993; 4.5-month-old, PMID: 36661119; neonatal diabetes median 4 weeks, PMID: 40887432), though nonsyndromic deafness may present in childhood. The disease is chronic, progressive, and lifelong with no spontaneous remission. The early-onset encephalopathic form sharing Alpers/Leigh features *"was characterized by more severe disease course and poorer survival compared to the other NARS2 associated phenotypes"* (PMID: 33596490). Severe cases include rapidly progressive intractable epilepsy with global brain atrophy and death around 9 months (*"focal status epilepticus that progressed to lethal epileptic encephalopathy"*, PMID: 30327238) and fatal refractory status epilepticus (PMID: 34415467). The mild end—isolated sensorineural hearing loss—is compatible with prolonged survival.

Finding 9 — Model systems & comparative biology

Human **NARS2** = NCBI Gene 79731; conserved orthologs include mouse *Nars2* (Gene 244141), rat *Nars2*, zebrafish *nars2*, and the essential yeast mitochondrial AsnRS ortholog *SLM5* (YCR024C). Published disease models are **cellular/in vitro**—patient-derived fibroblasts and EBV-transformed lymphoblasts showing decreased charged mt-tRNA-Asn, reduced OCR/ETC activity, and rescue by wild-type NARS2 (*"a decrease in oxygen consumption rates (OCR) and*

electron transport chain (ETC) activity can be rescued by overexpression of wild type NARS2", PMID: 25807530). High-throughput LC-MS/MS aminoacylation assays in patient fibroblasts classify variants ("developed a high-throughput LC-MS/MS-based aminoacylation assay to measure aaRS activity in patient-derived fibroblasts", PMID: 42028791). No dedicated knockout/knock-in animal model has been prominently reported, and no naturally occurring NARS2 disease is catalogued in OMIA; complete loss of the mitochondrial AsnRS is expected to be lethal, limiting constitutive-null organismal models.

Finding 10 — Etiology is purely monogenic; identifiers and prevention

COXPD24 is a Mendelian, autosomal-recessive disorder caused solely by biallelic NARS2 variants. There are **no established environmental, infectious, toxic, lifestyle, or dietary causal or protective factors**, and no somatic/acquired etiology (all variants are germline). The only genetic risk factor is inheritance of two damaging NARS2 alleles; consanguinity increases the risk of homozygosity. As in other mitochondrial disorders, intercurrent catabolic stressors (febrile illness, fasting) may precipitate metabolic decompensation (inferred from general mitochondrial biology, not NARS2-specific data). A representative statement: COXPD24 is "*a rare mitochondrial and a multisystem autosomal recessive disorder*" (PMID: 34374940). Prevention is limited to genetic counseling, cascade carrier testing, prenatal diagnosis, and preimplantation genetic testing (PGT-M) once familial variants are known.

Section-by-Section Report

1. Disease Information

COXPD24 is a rare autosomal-recessive multisystem mitochondrial disorder of oxidative phosphorylation caused by defective intramitochondrial translation. **Key identifiers:** MONDO:0014547; OMIM #616239 (Combined oxidative phosphorylation deficiency 24); gene NARS2 (OMIM 612803, HGNC:24406, NCBI Gene 79731, Ensembl ENSG00000137513, UniProt Q96I59); *allelic phenotype nonsyndromic deafness DFNB94*. Broad ontology mappings: ICD-10 E88.49/E88.40 (disorders of mitochondrial metabolism), ICD-11 5C53.1x (disorders of mitochondrial OXPHOS), MeSH grouping Mitochondrial Diseases (D028361). Synonyms: COXPD24; NARS2-related disease/disorder; mitochondrial asparaginyl-tRNA synthetase

*deficiency. Data source:** disease-level information is aggregated from individual patient reports and small case series consolidated in systematic reviews ([PMID: 33596490](#)), not from EHR-scale registries.

2. Etiology

Causal factor: purely genetic—biallelic pathogenic NARS2 variants. **Genetic risk:** two damaging NARS2 alleles; consanguinity elevates homozygosity risk; recurrent founder-type alleles (e.g., Scandinavian p.Pro214Leu). **Environmental risk/protective factors:** none established. **Gene–environment interaction:** none proven; catabolic stress may act as a nonspecific decompensation trigger (inferred).

3. Phenotypes

Phenotype	Type	Onset	Frequency / Notes	Suggested HPO
Sensorineural hearing loss	Clinical sign	Congenital/ childhood	Cardinal feature; isolated (DFNB94) or with CNS disease	HP:0000407
Refractory epilepsy / epileptic encephalopathy	Clinical sign	Infantile	Common in severe forms; epilepsy partialis continua reported	HP:0011097, HP:0200134
Global developmental delay / intellectual disability	Clinical sign	Infantile	Frequent	HP:0001263, HP:0001249
Leigh syndrome (symmetric basal ganglia lesions)	Imaging/ clinical	Infantile	Severe subgroup	HP:0002415
Alpers-like progressive encephalopathy / brain atrophy	Imaging/ clinical	Infantile	Severe, poor survival	HP:0002059, HP:0007303
Myopathy / hypotonia	Clinical sign	Infantile	Combined complex I/ IV deficiency on biopsy	HP:0003198, HP:0001252
Spastic paraplegia / ataxia	Clinical sign	Variable	Milder spectrum	HP:0001258, HP:0001251
Neonatal/insulin-dependent diabetes	Lab/clinical	Neonatal (median 4 wk)	Newly recognized; β -cell involvement	HP:0000857, HP:0100651
Lactic acidosis	Lab abnormality	Variable	Supports diagnosis	HP:0003128
Low birthweight	Physical	Congenital	7/8 in diabetes cohort	HP:0001518

Severity/progression: variable and genotype-graded, generally progressive; severe encephalopathic forms are life-limiting. **Quality-of-life impact:** major in severe forms (profound disability, seizures, sensory loss); disease-specific QoL instruments have not been applied.

4. Genetic/Molecular Information

Causal gene: NARS2 (11q14.1). **Variant classes:** predominantly missense and splice-site; also nonsense/truncating (p.Tyr323*). **Representative variants:** c.822G>C splice, p.Val213Phe, p.Asn381Ser, p.Tyr323*, p.Pro214Leu, p.Ile182Lys, p.His167Arg, p.Phe216Leu, p.Thr180Asn, p.Val440Leu. **Population frequency:** disease alleles are ultra-rare; gnomAD shows NARS2 is LoF-tolerant (pLI $\approx$ 0, oe_lof $\approx$ 0.95), consistent with recessive inheritance. **Origin:** germline (no somatic disease). **Functional consequence:** loss of function via reduced aminoacylation, impaired dimerization (p.Asn381Ser), or absent protein (p.Tyr323*). **Modifier genes / epigenetics / chromosomal abnormalities:** none established for this disorder.

5. Environmental Information

No environmental, occupational, toxic, lifestyle, or infectious factors are implicated. Not applicable as a causal or protective category for this monogenic disease.

6. Mechanism / Pathophysiology

Ordered causal chain: 1. Biallelic NARS2 variants **lead to** a structurally/functionally defective mitochondrial asparaginyl-tRNA synthetase (loss of function; reduced dimerization or absent protein). 2. Defective mt-AsnRS **results in** reduced aminoacylation (charging) of mitochondrial tRNA-Asn (directly demonstrated: decreased charged mt-tRNA-Asn). 3. Reduced charged mt-tRNA-Asn **impairs** mitochondrial translation of the 13 mtDNA-encoded OXPHOS subunits. 4. Impaired mitochondrial translation **causes** a combined OXPHOS assembly/activity defect, predominantly of complex I and complex IV (with complex V subcomplexes on BN-PAGE). 5. Combined OXPHOS deficiency **leads to** reduced oxygen consumption and ATP production (reduced OCR/ETC activity; rescuable by wild-type NARS2). 6. Bioenergetic failure **results in** cell/tissue dysfunction in high-energy-demand tissues, branching to: - CNS neurons $\rightarrow$ Leigh/Alpers encephalopathy, epilepsy, developmental regression (bilateral symmetric basal ganglia/thalamus/brainstem, or cortical atrophy). - Cochlear/inner-ear cells $\rightarrow$ sensorineural hearing loss. - Skeletal myocytes $\rightarrow$ myopathy/hypotonia. - Pancreatic β -cells $\rightarrow$ impaired insulin

secretion → neonatal diabetes (inferred mechanism from β -cell energy dependence). 7. Progressive tissue injury **culminates in** the clinical manifestations and, in severe forms, early death.

Molecular/cellular detail: biological process—mitochondrial translation (GO:0032543), asparaginyl-tRNA aminoacylation (GO:0006421); cellular component—mitochondrial matrix (GO:0005759). Cell types (CL): neuron (CL:0000540), cochlear hair cell (CL:0000589/CL:0002494), skeletal muscle cell (CL:0000188), pancreatic beta cell (CL:0000169). Metabolic consequence: energy-metabolism failure with lactic acidosis (CHEBI:24996 lactate). Immune involvement: none characteristic.

7. Anatomical Structures Affected

Primary organs/systems: central nervous system (UBERON:0000955 brain; basal ganglia UBERON:0002420; thalamus UBERON:0001897; brainstem UBERON:0002298), inner ear/cochlea (UBERON:0001844), skeletal muscle (UBERON:0001134), endocrine pancreas/islets (UBERON:0000006). **Subcellular:** mitochondrial matrix (GO:0005759). **Laterality:** CNS and cochlear involvement typically bilateral/symmetric.

8. Temporal Development

Onset: congenital to early childhood; most severe cases neonatal–infantile. **Pattern:** subacute-to-chronic, progressive; may show acute decompensations (status epilepticus, stroke-like episodes). **Stages/course:** progressive neurodegeneration in severe forms; stable/slowly progressive in isolated deafness. **Duration:** lifelong. **Remission:** none spontaneous. **Critical period:** early infancy is the key window for diagnosis and supportive intervention.

9. Inheritance and Population

Inheritance: autosomal recessive. **Epidemiology:** ultra-rare; precise prevalence/incidence not established (fewer than ~40 reported patients). **Penetrance:** high for biallelic damaging genotypes; **expressivity** highly variable and genotype-dependent. **Anticipation:** not applicable (not a repeat-expansion disorder). **Founder effects/consanguinity:** recurrent homozygous alleles in consanguineous populations (Scandinavian, Iranian, Turkish). **Carrier frequency:** low; heterozygous carriers are unaffected (LoF-tolerant gene). **Sex ratio:** no sex bias (autosomal). **Geographic distribution:** worldwide, individual families.

10. Diagnostics

First-line: whole-exome or whole-genome sequencing (trio-based preferred). **Supportive labs:** elevated blood/CSF lactate, myocardial/muscle enzymes. **Imaging:** MRI (Leigh-type symmetric lesions; Alpers-like atrophy; delayed myelination). **Electrophysiology:** EEG (epileptiform activity, epilepsy partialis continua). **Muscle biopsy:** combined complex I + IV deficiency (spectrophotometry, BN-PAGE). **Functional/omics:** mt-tRNA-Asn charging assays, wild-type rescue, LC-MS/MS aminoacylation for VUS resolution. **Differential diagnosis:** other mt-ARS disorders (AARS2, DARS2, YARS2, IARS2), other Leigh/Alpers etiologies, and other causes of nonsyndromic deafness or neonatal diabetes. **Screening:** cascade carrier testing of relatives; no newborn screening.

11. Outcome / Prognosis

Prognosis is **genotype-linked and often severe**. Early-onset Leigh/Alpers-type disease has poor survival with high early mortality; isolated sensorineural hearing loss is compatible with normal lifespan. **Morbidity** in severe forms is profound (refractory epilepsy, developmental disability, sensory loss, insulin-dependent diabetes). **Prognostic factors:** age of onset, encephalopathic phenotype, and structural/biochemical severity of the variant. No validated prognostic biomarkers beyond genotype and lactate.

12. Treatment

Supportive only: antiepileptic drugs (avoid/limit valproate; NCIT anticonvulsant therapy), hearing rehabilitation (hearing aids, cochlear implants; NCIT), developmental/physical/occupational/speech therapy, nutritional support, and insulin for the neonatal-diabetes subgroup (NCIT insulin therapy). **Experimental:** L-asparagine (CHEBI:22653) substrate repletion—rational but unproven; class-level amino-acid supplementation trials in mt-ARS leukoencephalopathies were safe but ineffective ([PMID: 41075682](#)). **Advanced therapeutics:** no approved gene, cell, or RNA therapy; gene therapy for ARS disorders is being explored mainly for dominant neuropathies. **Pharmacogenomics:** valproate caution in mitochondrial disease.

13. Prevention

No primary prevention (no vaccine, no modifiable risk factor). **Secondary/tertiary:** early diagnosis and supportive management of complications. **Genetic prevention:** genetic counseling, cascade carrier testing, prenatal diagnosis, and preimplantation genetic testing (PGT-M) once familial variants are identified.

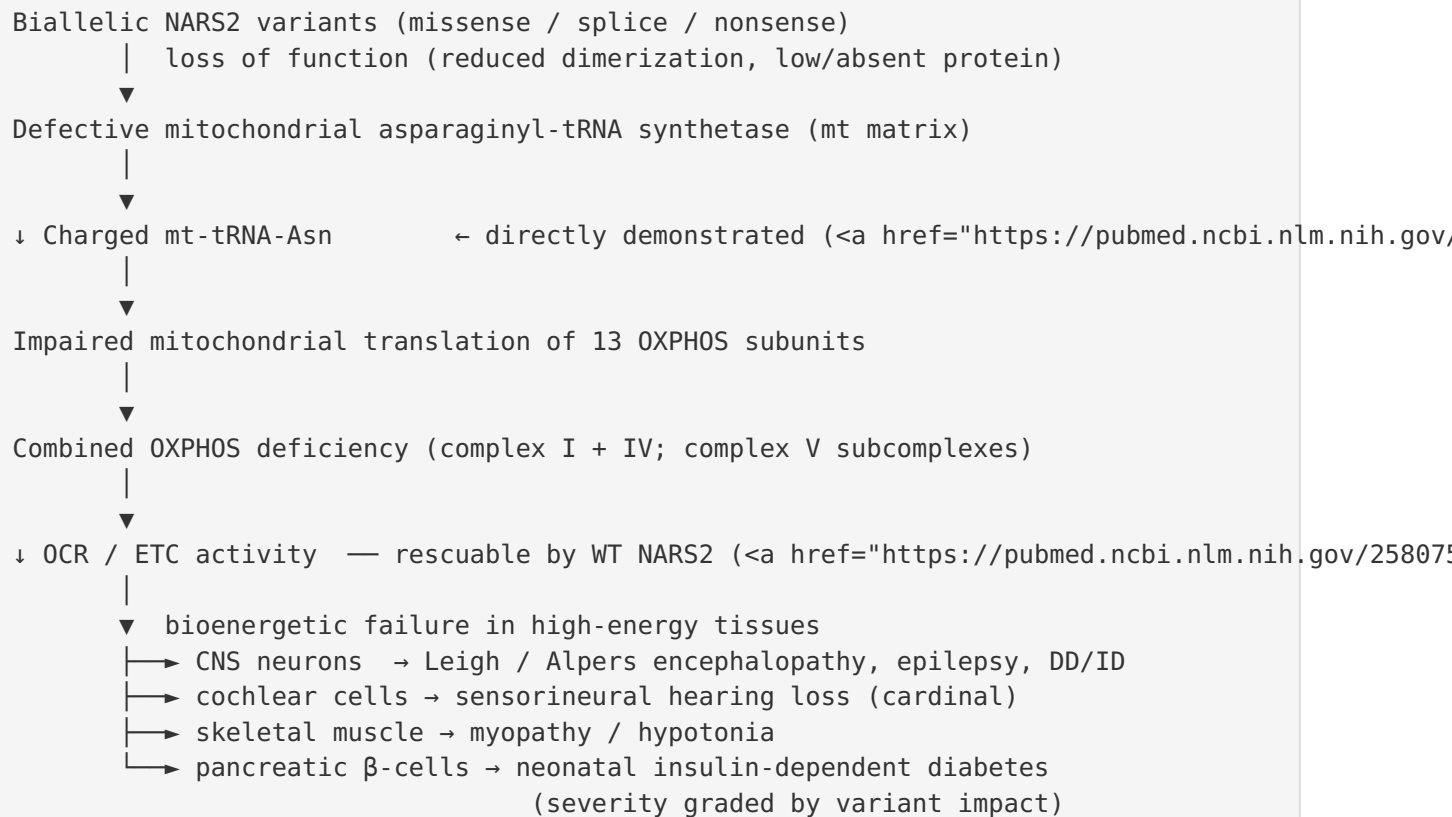
14. Other Species / Natural Disease

Conserved orthologs: mouse *Nars2* (NCBI Gene 244141), rat *Nars2*, zebrafish *nars2*, yeast *SLM5* (essential mitochondrial AsnRS). No naturally occurring NARS2 disease is catalogued in companion animals or wildlife (OMIA). No zoonotic relevance. The mitochondrial translation mechanism is evolutionarily conserved, but organismal disease has not been reported outside humans.

15. Model Organisms

Established models are **cellular/in vitro**: patient-derived skin fibroblasts and EBV-transformed lymphoblasts recapitulating decreased charged mt-tRNA-Asn, reduced OCR/ETC activity, and complementation by wild-type NARS2. High-throughput LC-MS/MS aminoacylation assays in patient fibroblasts serve variant classification. **Limitation**: no dedicated knockout/knock-in mouse, zebrafish, or fly disease model is prominently reported; constitutive nulls are expected to be embryonic-lethal, favoring conditional/hypomorphic or iPSC-derived approaches for future work.

Mechanistic Model / Interpretation



The disease is a textbook example of a **mitochondrial translation defect**: a single enzymatic step (Asn-tRNA charging) constrains synthesis of the mtDNA-encoded OXPHOS subunits, producing a **combined** rather than isolated respiratory-chain deficiency. The consistency of the genotype–phenotype correlation—structural severity predicting clinical severity—supports a **dosage/residual-activity model** in which hypomorphic alleles yield tissue-restricted phenotypes (e.g., deafness only) while more damaging biallelic combinations yield multisystem encephalopathy.

Evidence Base

PMID	Title (abbrev.)	Contribution
25385316	Homozygous splice-site NARS2, two siblings	Establishes combined complex I + IV deficiency and reduced charged mt-tRNA-Asn
25807530	NARS2 → deafness & Leigh syndrome	Gene-disease proof; complementation rescue; genotype–phenotype correlation
33596490	Phenotypic variability & natural history	Hearing loss cardinal; early-onset form has poorer survival
40887432	Neonatal diabetes in COXPD-24	Expands phenotype; β-cell involvement
34374940	Novel phenotype/genotype spectrum	Confirms OMIM ID, AR multisystem nature
36661119	Turkish variant with T1DM	Valproate caution; WES shortens diagnosis
30327238	Lethal rapidly progressive epilepsy	Rapid WES diagnosis; fatal course
34415467	Fatal refractory status epilepticus	Severe end of spectrum
41426993	Stroke-like lesion & status epilepticus	Infantile onset; lactate, MRI features
36523694	Nonsyndromic SNHL from NARS2	Confirms DFNB94 phenotype
42028791	aaRS variant classification & therapy	Functional assays; therapeutic landscape
41075682	Amino-acid supplementation trial (mt-ARS)	Safe but no efficacy—class-level therapy evidence
41404429	Tyrosine supplementation in YARS2/MLASA2	Modest gains in select cases (analogy)

Evidence is predominantly **human clinical** (case reports/series and one systematic review) plus **in vitro** functional studies in patient-derived cells. There are no randomized trials, and no NARS2-specific therapeutic trial exists.

Limitations and Knowledge Gaps

- **Small sample size:** fewer than ~40 patients reported worldwide; prevalence/incidence are unknown and phenotype frequencies are largely qualitative.
 - **No animal model:** no established knockout/knock-in organismal model limits mechanistic and preclinical therapeutic work; constitutive nulls are likely lethal.
 - **Genotype–phenotype resolution incomplete:** while structural severity correlates with clinical severity, the basis of tissue-specificity (e.g., why some alleles cause deafness only) is not fully explained.
 - **β -cell mechanism inferred:** the pathway from OXPHOS deficiency to impaired insulin secretion is plausible but not directly demonstrated in NARS2 β -cells.
 - **No proven therapy:** L-asparagine repletion is untested in NARS2; class-level supplementation trials were negative.
 - **No modifier genes, epigenetic, or environmental data** are available for this disorder.
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Proposed Follow-up Experiments / Actions

1. **Establish a natural-history registry** aggregating all reported and unpublished cases to quantify phenotype frequencies, survival, and genotype–phenotype correlations.
2. **Generate iPSC-derived models** (neurons, cochlear-like organoids, β -cells) from patients to test tissue-specific bioenergetic phenotypes and the β -cell insulin-secretion defect directly.
3. **Develop conditional/hypomorphic mouse or zebrafish models** to enable in vivo mechanistic and preclinical therapeutic studies without constitutive lethality.
4. **Test L-asparagine substrate repletion** in patient-derived cells (aminoacylation, OCR/ETC rescue) as a rational precursor to any clinical evaluation.
5. **Standardize variant functional classification** using the LC-MS/MS aminoacylation assay to reclassify VUS and refine genotype–phenotype models.
6. **Explore AAV/gene-replacement feasibility** for NARS2 given its recessive loss-of-function mechanism and successful in vitro complementation.
7. **Confirm ontology/identifier mappings** (ICD-11 5C53.1x, MONDO:0014547) and add HPO frequency annotations to the knowledge base entry.

Evidence source types: human clinical (case reports/series, systematic review), in vitro (patient fibroblasts/lymphoblasts, complementation, LC-MS/MS assays), and computational (gnomAD constraint, UniProt/ontology annotation). Citations use PubMed IDs (PMIDs). Where information was unavailable for this ultra-rare disorder, this is stated explicitly.

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